SSL Playbook — b2bwebapp.com

Unified Knowledge Base & Runbook for Certificate Installation, Debugging, and Automation.

Overview

This document is a complete single-source SSL playbook containing: • End-to-end steps from CSR creation to installation in Kubernetes. • Debugging steps for all common SSL/Ingress issues. • Lessons learned from real production incident. • cert-manager automation setup. • Automated CI-based secure rotation pipeline. • One-page on-call checklist.

Prerequisites

• Access to DNS provider (GoDaddy). • Access to Azure Kubernetes Cluster (AKS). • kubectl configured with prod namespace. • CA-issued certificate files (leaf + intermediate). • Private key stored securely.

Files received from CA

Usually: • leaf.crt • intermediate.crt • fullchain.crt Full chain must ALWAYS be: leaf first → intermediates next.

Creating Key + CSR

openssl genpkey -algorithm RSA -out b2b.key -pkeyopt rsa_keygen_bits:2048 openssl req -new -key b2b.key -out b2b.csr -config csr.conf CSR SAN config: [req] req_extensions = v3_req distinguished_name = req_distinguished_name [req_distinguished_name] [v3_req] subjectAltName = @alt_names [alt_names] DNS.1 = b2bwebapp.com DNS.2 = *.b2bwebapp.com DNS.3 = app.b2bwebapp.com DNS.4 = api.b2bwebapp.com

Assemble Fullchain

cat leaf.crt intermediate.crt > b2b.fullchain.crt grep -c 'BEGIN CERTIFICATE' b2b.fullchain.crt

Create TLS Secret in prod

kubectl -n prod create secret tls b2b-tls --cert=b2b.fullchain.crt --key=b2b.key --dry-run=client -o yaml | kubectl apply -f -

Ingress TLS Reference

spec: tls: - hosts: - app.b2bwebapp.com - api.b2bwebapp.com secretName: b2b-tls

Restart Ingress Controller

kubectl -n ingress-nginx rollout restart deployment ingress-nginx-controller

Verify Secret

kubectl -n prod get secret b2b-tls -o jsonpath='{.data.tls\.crt}' | base64 --decode | openssl x509 -noout -subject -issuer -dates -ext subjectAltName

Check Logs

POD=$(kubectl -n ingress-nginx get pods -l app.kubernetes.io/component=controller -o jsonpath='{.items[0].metadata.name}') kubectl -n ingress-nginx logs $POD -c

```
controller --tail=300 | egrep -i 'Adding secret|no valid PEM|fake certificate'
```

Test SNI

```
openssl s_client -connect :443 -servername app.b2bwebapp.com | openssl x509
-noout -subject -issuer
```

Curl Test

```
curl -v --resolve app.b2bwebapp.com:443: https://app.b2bwebapp.com/
```

Common Issues & Fixes

```
1. Fake Certificate: Cause: secret mismatch or bad PEM. Fix: recreate secret &
restart. 2. no valid PEM formatted block: Cause: corrupted cert or wrong order.
Fix: rebuild fullchain. 3. LB serving old cert: Cause: controller not reloaded.
Fix: restart ingress controller. 4. Azure Front Door in path: Must upload cert
there too.
```

Manual Renewal Steps

```
kubectl -n prod create secret tls b2b-tls --cert=new_fullchain.crt
--key=new.key   --dry-run=client -o yaml | kubectl apply -f - kubectl -n
ingress-nginx rollout restart deployment ingress-nginx-controller
```

cert-manager Automation

```
apiVersion: cert-manager.io/v1 kind: Certificate metadata:   name: b2b-cert
  namespace: prod spec:   secretName: b2b-tls   dnsNames:   - app.b2bwebapp.com
  - api.b2bwebapp.com   issuerRef:     name: letsencrypt-staging     kind:
ClusterIssuer
```

CI/CD Rotation Pipeline

```
Steps: 1. Download cert from CA or Vault. 2. Create TLS secret: kubectl -n prod
create secret tls b2b-tls --cert=cert.pem --key=key.pem --dry-run=client -o yaml
| kubectl apply -f - 3. Restart controller. 4. Smoke test using openssl and curl.
```

On-call Checklist

```
1. dig app.b2bwebapp.com. 2. kubectl -n prod get secret b2b-tls. 3. Decode cert →
verify expiry & SANs. 4. Check ingress secretName. 5. Check logs for PEM errors.
6. openssl SNI test. 7. curl test. 8. Fix → recreate secret & restart. 9. Record
new expiry.
```

End of SSL Playbook — b2bwebapp.com